

ShadowCore-RV: A Custom Fault-Tolerant RV32I Processor

Track 1 - FPGA

Application Domain - Processor

PS-01: Fault-Tolerant Processor for Harsh Environments

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ABSTRACT

As semiconductor scaling continues, microprocessors operating in challenging environments become increasingly susceptible to transient faults and soft errors. While standard single error correction, double error detection (SECDDED) codes are commonly implemented to mitigate these risks, they remain vulnerable to syndrome aliasing, where multi-bit errors are misclassified and silently corrupt the architectural state. This proposal introduces a novel, fault-tolerant processor core architecture based on the RISC-V instruction set [2]. The proposed design implements a comprehensive Triple-Axis Validation system. For the data path, it combines Hamming SECDDED with a parallel Cyclic Redundancy Check (CRC) to confidently validate single-bit corrections and reject aliased multi-bit faults. To ensure holistic reliability, this is augmented with spatial and semantic bound checks for the address path, and an asymmetric dual-decoder for control signal verification. Detection logic executes in parallel with the standard pipeline, converging at a novel Writeback (WB) Commitment Firewall. This barrier prevents corrupted data from entering the register file, triggering a lightweight, 1-cycle REDO or selective FLUSH recovery mechanism rather than relying on standard pipeline stalls. This architecture aims to provide highly robust, real-time transient fault mitigation with optimized FPGA resource utilization.

Keywords: processor architecture; RISC-V; fault-tolerant; SECDDED; cyclic redundancy check; pipeline recovery; hardware reliability

MAIN IDEA

The proposed Hybrid Fault-Tolerant RV32I Processor addresses the critical trade-off between reliability and area overhead in edge environments. While foundational microarchitectural concepts are adapted from the baseline DuckCore processor [1], this design introduces a novel distributed validation paradigm to eliminate silent data corruption (SDC). It achieves this without incurring the severe Look-Up Table (LUT), Block RAM, and routing penalties associated with full-datapath Triple Modular Redundancy (TMR). The architecture is defined by four core hardware subsystems:

1. Distributed “Shadow Check” Pipeline (Data Integrity)

To maintain a base CPI of 1, error detection is decentralized and operates concurrently with standard pipeline transitions (IF/ID, ID/EX, EX/MEM, MEM/WB).

- **Dual-Layer Data Verification:** Pipeline registers encapsulate the 32-bit data payload alongside a 7-bit Hamming SECDED syndrome and a parallel hardware Cyclic Redundancy Check (e.g., CRC-8).
- **Combinational Validation Flow:** The SECDED logic proposes a 1-bit correction candidate via combinational XOR trees. Simultaneously, the CRC module recalculates the checksum of this candidate to act as an independent validator.
- **Deterministic Aliasing Rejection:** If the CRC validation yields a PASS, the single-bit correction is committed locally without stalling the pipeline. A CRC FAIL definitively identifies multi-bit aliasing, flagging the payload as UNCORRECTABLE and preventing corrupt architectural state propagation.

2. Asymmetric Control-Flow Integrity (CFI)

Because the Program Counter (PC) dictates the execution state, it requires an isolated, rigorous protection scheme independent of the data buses.

- **Localized PC TMR:** The physical PC register is triplicated. A continuous hardware majority voter resolves single-event upsets (SEUs) instantly, masking transient faults before they affect the instruction fetch stage.
- **Deterministic CFI Watchdog:** A dedicated hardware monitor continuously asserts `Actual_PC == Expected_PC` (derived from sequential `PC+4` or validated branch targets) to trap erroneous control-flow divergences.
- **Semantic Bound Enforcement:** Zero-latency hardware assertions enforce physical LSB alignment (ensuring $PC \pmod{4} = 0$) and strictly bound execution to valid memory address spaces.

3. Decoupled Control Signal Protection

Protecting control signals with full SECDED and CRC creates unacceptable timing delays and area overhead. This architecture deliberately decouples control from data protection.

- **Lightweight Parity Trees:** Critical control vectors (e.g., `RegWrite`, `MemRead`, `MemWrite`, `Branch`) are protected via fast, lightweight parity bits.
- **Pre-Emptive Error Flagging:** A parity mismatch instantly asserts an error flag to the central arbiter, nullifying the control signal before it can trigger an invalid memory or register operation.

4. The Writeback (WB) Barrier & ARBIT State Machine

The physical Writeback stage acts as a strict containment barrier, isolating unverified data from the processor’s permanent architectural state.

- **Gated Commitment:** Architectural writeback is strictly gated by the combinational logic: `safe_regwrite = normal_regwrite && wb_valid`.
- **Centralized Recovery (ARBIT):** Localized error supervisors feed status flags into the ARBIT finite state machine. While valid instructions commit normally, an `UNCORRECTABLE` flag pulls `wb_valid` low and triggers a 1-cycle REDO from a lightweight pipeline checkpoint.
- **Selective Rollback & Fault Escalation:** Errors on instructions already invalidated by branching logic trigger a standard FLUSH rather than a rollback. Persistent faults increment a hardware counter, eventually escalating to a `PERSISTENT` flag that forces the core into a safe reset state.

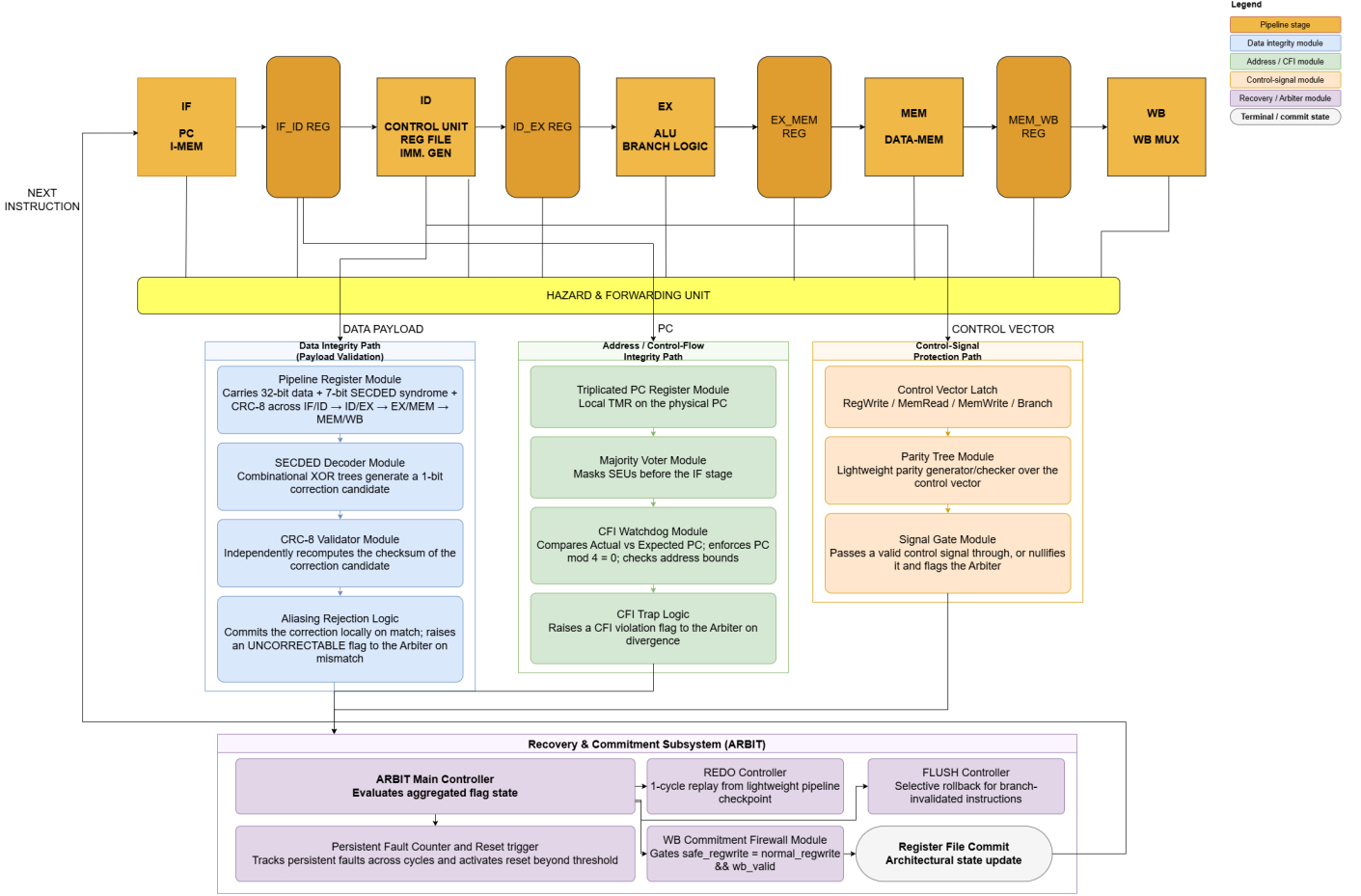


Figure 1: System-level block diagram of the ShadowCore-RV Processor, illustrating the parallel execution of the Triple-Axis Validation logic and the ARBIT containment state machine.

APPLICATION

The proposed Triple-Axis Validation RISC-V architecture is highly suited for mission-critical and safety-critical domains where environmental factors increase the likelihood of transient hardware faults (soft errors). Potential and future applications include:

- Aerospace and Deep-Space Exploration:** Satellites, lunar/Martian rovers, and deep-space probes are constantly exposed to high levels of cosmic radiation and heavy ions, which cause Single Event Upsets (SEUs). The architecture's ability to catch multi-bit aliasing without halting the system ensures uninterrupted operation in environments where physical repair is impossible.
- Automotive and Autonomous Vehicles (ADAS):** Modern vehicles rely on complex System-on-Chips (SoCs) for steering, braking, and navigation. To meet strict ISO 26262 functional safety standards, processors must mitigate random hardware failures. Our lightweight 1-cycle REDO mechanism offers real-time recovery, preventing catastrophic delays that standard software-level reboots would cause at highway speeds.

- **High-Altitude Aviation:** Neutron flux increases by nearly 100x at commercial cruising altitudes (30,000+ feet) compared to sea level. Flight control systems and avionics can utilize this core to ensure transient faults do not corrupt navigation data or autopilot control signals.
- **Industrial Automation and Heavy Machinery:** Manufacturing plants, foundries, and automated robotics operate in environments with severe Electromagnetic Interference (EMI) and power grid fluctuations. The Triple-Axis protection ensures that control wires and memory states remain uncorrupted by localized electrical noise.
- **Extreme Climate Edge Computing (Harsh Weather):** Remote IoT sensors, early-warning weather stations, and edge servers deployed in extreme environments (e.g., deserts, arctic regions) face severe thermal fluctuations. Extreme temperatures can increase semiconductor susceptibility to soft errors and timing faults, making self-correcting hardware vital for remote maintenance.
- **Critical Medical Devices:** Life-supporting equipment (e.g., dialysis machines, surgical robots) and radiology devices (e.g., CT/PET scanners) operate in proximity to localized radiation or require zero-downtime reliability. The Writeback Commitment Firewall ensures that no corrupted data accidentally alters patient treatment parameters.

VALUE ADD

This proposed architecture provides a highly innovative, resource-conscious approach to processor reliability, offering distinct advantages over traditional fault-tolerant implementations:

- **Custom Microarchitectural Foundation:** Rather than wrapping a black-box open-source IP, this fault-tolerant overlay is integrated into a custom-designed, from-scratch RV32I base core. This deep integration allows for cycle-accurate manipulation of the pipeline registers and the ARBIT state machine, ensuring optimal SystemVerilog synthesis and avoiding unnecessary routing bloat on the FPGA.
- **Novel Solution to Syndrome Aliasing (Innovation & Originality):** Standard SECDED architectures are vulnerable to multi-bit errors that alias to single-bit syndromes, leading to silent data corruption. Our hybrid Hamming+CRC logic explicitly solves this by adopting a "correct when confident; recover when uncertain" paradigm, catching aliased faults before they commit.
- **Comprehensive Triple-Axis Protection:** Unlike conventional designs that only protect memory payloads, this architecture validates all critical vectors: Data (SECDED+CRC), Address (spatial bounds and PC alignment), and Control (asymmetric dual-decoding).
- **Optimized Hardware Resource Utilization:** The design deliberately avoids the massive LUT, Flip-Flop, and power overhead associated with Triple Modular Redundancy (TMR). Utilizing an asymmetric micro-decoder for control wires and parallel XOR trees for the CRC ensures maximum protection with minimal silicon footprint.
- **High Throughput via Hardware Parallelism:** The fault-detection logic executes concurrently with the standard IF/ID/EX pipeline stages. This parallel integration guarantees zero processing cycle overhead during normal operations, maintaining optimal throughput and a base CPI (Cycles Per Instruction) of 1.
- **Ultra-Low Latency Recovery:** By centralizing validation at the Writeback (WB) Commitment Firewall, the architecture isolates corrupted states before architectural commit. The ARBIT supervisor then triggers a lightweight, 1-cycle REDO (replay) mechanism, significantly reducing the recovery latency compared to traditional full-pipeline flushes or software-level traps.

REFERENCES

- [1] J. Li, S. Zhang, and C. Bao, “DuckCore: A Fault-Tolerant Processor Core Architecture Based on the RISC-V ISA,” *Electronics*, vol. 11, no. 1, Art. no. 122, 2022, doi: 10.3390/electronics11010122.
- [2] RISC-V International, *The RISC-V Instruction Set Manual, Volume I: Unprivileged Architecture, RV32I Base Integer Instruction Set*, Version 2.1, 2024.